

UGW9811

V900R019C10

5G NSA Subscriber Information Report Feature Guide

1 Before You Start

Table1 As defined in 3GPP specifications, the non-standalone (NSA) network leverages existing network resources and allows 4G eNodeBs and 5G gNodeBs to be connected to the same EPC or 5GC, accelerating evolution from 4G to 5G. In the early stage of 5G deployment, NSA allows control data to be transmitted from the 5G RAN to the EPC through the 4G RAN, so that 5G RANs can be deployed for commercial sites even if 5GC specifications have not been frozen. This makes full use of existing 4G network resources and accelerates time to market (TTM) of 5G. In large-scaled 5G deployment, NSA helps transmit control data from the 4G RAN to the 5GC through the 5G RAN, making it possible for carriers to decommission 4G networks.

Table2 The MME, S-GW, and P-GW on the NSA network are required to work together to distinguish 4G traffic from 5G traffic for 5G NSA UEs.

Table3

2 5G NSA Subscriber Information Report Feature Guide

Table4 **Applicable NEs**

Table5 S-GW and P-GW

Table6 **Definition**

Table7 The 5G NSA Subscriber Information Report feature is designed to differentiate 4G traffic from 5G NSA traffic. The MME dynamically learns eNodeB addresses, automatically maintains the eNodeB address list, and identifies 5G NSA UEs based on the list. The MME sends a proprietary extension IE to notify the UGW9811 functioning as an S-GW or P-GW that a UE is located on a 4G or 5G network. The P-GW sends an RAT type change event to the OCS server/PCRF to charge 5G NSA services separately.

Table8 **NOTE**

Table9 This is a customized feature. If it does not meet commercial use requirements, submit a requirement for customization.

Table10 **Benefits**

Table11 For...	Table12 Benefits
Carriers	Carriers can accurately identify 5G NSA UEs, design charging packages separately, and maximize the return on investment (ROI).
Subscribers	Subscribers do not need to be aware of this feature.

Table13 **Availability**

Table14 **Involved NEs**

Table15 NE	Table16 Supported Version	Table17 Function
S-GW/P-GW	◆ CloudUGW V100R020C60SP H125 and later ◆ UGW9811 V900R019C10SP H159 and later	◆ Identifies 5G NSA UEs based on information about the UGW9811 that serves them. ◆ Sends a specific NAT type to the PCRF/OCS server/CG for 5G NSA UEs.
MME	◆ CloudUSN V100R020C60S	◆ Identifies UEs that are served by the gNodeB and sends the

Table15 NE	Table16 Supported Version	Table17 Function
	PH129 and later ◆ USN9810 V900R019C10S PH317 and later	UE information to the UGW9811.
PCRF (optional)	Versions that can identify the new RAT value (10)	◆ Manages charging policies or bandwidth based on the RAT type of 5G NSA UEs.
OCS server (optional)	Versions that can identify the new RAT value (10)	◆ Performs online charging for 5G NSA UEs based on the RAT type.
DRA (optional)	Versions that can identify the new RAT value (10)	◆ Forwards Diameter messages.
AAA server (optional)	Versions that can identify the new RAT value (10)	◆ The AAA accounting server performs accounting based on the RAT type of NSA UEs.
CG (optional)	◆ CG9812 V500R006C20SP C305 and later ◆ CloudCG V100R019C15SP C500 and later	◆ Performs offline charging for 5G NSA UEs based on the RAT type.

Table18

Table19 **Requirement for the License**

Table20 This feature is customized.

Table21 **Interaction with Other Features**

Table22 Category	Table23 Feature	Table24 License Control Item	Table25 Description
Dependancy	GWFD-110410 Supporting 5G NSA Networking	◆ 82209119 LKW5W5GSPG 01 5G SGW&PGW Basic Software for UeMBB Slicing(per PDP) ◆ 82209120 LKW5W5GSGW 01 5G SGW Basic Software for UeMBB Slicing(per PDP) ◆ 82209121 LKW5W5GPGW 01 5G PGW Basic Software for UeMBB Slicing(per PDP)	The Supporting 5G NSA Networking feature must be activated because the UGW9811 can only charge 5G NSA subscribers who have accessed the 5G NSA network.
Dependancy	GWFD-110302 Basic Content Based Charging	N/A	The Basic Content Based Charging feature must be activated because the UGW9811 can only charge 5G NSA subscribers after identifying their services.

Table26

Table27 **Impacts on the System**Table28 **System Capacity**

Table29 None.

Table30 **Application Limitations**

- This feature involves customized private IEs and can be used only by Huawei MME/S-GW/P-GW. Otherwise, services on the local VNF and peripheral VNFs may be abnormal.
- The CG/AAA/PCRF/OCS should identify the extended RAT type (extended RAT type). Confirm that services will not be affected if the RAT type is added for the interworking between the CG/AAA/PCRF/OCS/DRA and peripheral VNFs.
- This feature does not take effect when the eNodeB and gNodeB on the RAN side use the same user-plane IP address.
- After this feature is enabled, it shall take a period of time to accurately identify UE types. The MME needs to learn eNodeB IP addresses based on the eNodeB user-plane IP address carried in the Initial Context Setup Response message. Currently, the feature takes effect **one day** after the feature is enabled.
- This feature cannot be used for NSA UEs that are performing services when this feature is enabled. The UEs can be identified only after they enter the Active state again.
- If the eNodeB is disconnected for more than three days, the data-plane IP address of the eNodeB stored in the system will be aged. In this case, the system needs to learn the data-plane IP address of the eNodeB contained in the Initial Context Setup Response message in the Attach, TAU, and Service Request procedures again.
- When an NSA-capable UE accesses the network through one of attach, TAU, X2-based handover, and S1-based handover procedures and the UE is within the coverage of a 4G base station, the Create Session Request message sent to the S-GW/P-GW carries the DCNR flag. However, because the proprietary IE can be transmitted to the P-GW through only a Modify Bearer Request message. During this period, 5G PGW-CDRs may be generated.
- During an S1-based handover procedure, the ULI IE in the Create Session Request message carries only the TAI but not the ECGI.
- If handover or location update frequently occurs for 5G NSA UEs, there is a higher probability that signaling conflicts occur on the S-GW/P-GW. To resolve the deadlock caused by the inter-NE signaling conflicts, the Modify Beater Request message carrying 4G/5G network changes may be optimized or discarded. As a result, 4G/5G network changes of a small number of 5G NSA UEs may be reported to peripheral NEs (PCRF/OCS server/CloudCG/AAA server/DRA) later than expected.
- After this feature is enabled, if the customer is concerned about the consistency between the types of the RAT and ULI in a CDR, you are advised to set **BIT922** to **0** so that the ULI filled in a CDR remains unchanged when the CDR is opened and is the same as the RAT.

Table31 **Implementation Principles**

Table32 **System Implementation**

Table33 The MME dynamically learns eNodeB addresses and automatically maintains the eNodeB address list.

When a UE moves between a 4G network and a 5G network, the MME checks whether the (R)AN address is in the eNodeB address list. If the (R)AN address is not in the eNodeB address list, the MME considers that the UE is located on a 5G network. The MME notifies the S-GW or P-GW of the UE location on the 5G network. The P-GW sends an RAT type change event to the OCS server/PCRF so that the OCS server/PCRF can charge 5G NSA services separately.

Table34 **Service Procedure**

Figure 1-1 5G NSA charging process

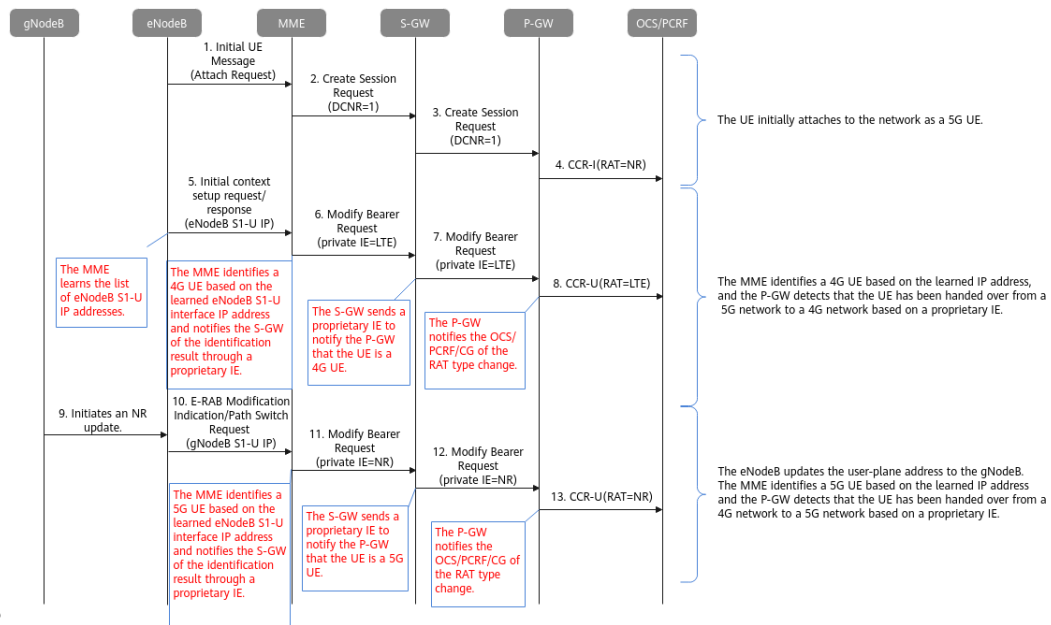


Figure2

Table35

Table36 Steps 1-4. A UE initially attaches to the network as a 5G UE.

Table37 Step 5. After receiving an Initial context setup request message, the eNodeB assigns an S1-U interface address and sends it to the MME through an Initial Context Setup Response message. The MME learns the user-plane IP address of the eNodeB from the response message.

Table38 Steps 6-8. The MME identifies a 4G UE based on the learned IP address, and the P-GW detects that the UE has been handed over from a 5G network to a 4G network based on a proprietary IE.

Table39 Steps 9-12. The eNodeB updates the user-plane address to the gNodeB. The MME identifies a 5G UE based on the learned IP address and the P-GW detects that the UE has been handed over from a 4G network to a 5G network based on a proprietary IE.

Table40 Step 13. If the PGW-C receives a new RAT (NR) and performs offline charging, it populates the customized RAT in CDRs to indicate NSA. If the PGW-C receives a customized RAT and performs online charging, it sends a notify RAT (NR) change event to the OCS server and PCRF.

Figure3

Table41 **Feature Specifications**

Table42 N/A

Table43 **Standards Compliance**

Table44 N/A

Table45 **Release History**

Table 1-1 Release history of the 5G NSA Subscriber Information Report feature

Table46 Feature Version	Table47 Product Version	Table48 Details
02	V900R019C10SPH159	Updated the NE versions and restrictions.
01	V900R019C10	First release.

Table49

1.1 Activating the 5G NSA Subscriber Information Report Feature

Table50 **Scenarios**

Table51 Activate the 5G NSA Subscriber Information Report feature on the UGW9811 if carriers plan to charge 5G NSA subscribers separately.

Table52  **NOTE**

Table53 This feature is used with the S-GW and P-GW.

Table54 **Prerequisites**

Table55 **Conditions**

Table56 5G NSA subscribers have been activated.

Table57 **Data**

Table 1-2 Data planned on the UGW9811

Table58 Category	Table59 Parameter	Table60 Example Value	Table61 How to Obtain	Table62 Description
5G NSA subscriber identification	BIT296 Controlling Whether the UGW9811 Updates the DCNR Flag During a Handover	1	Planned on the UGW9811	This software parameter is configured using the following command: softpara bit 296 1

	BIT327 Controlling Whether the UGW9811 Includes or Processes a DCNR or NR Flag in a Modify Bearer Request Message	1	Planned on the UGW9811	This software parameter is configured using the following command: softpara bit 327 1
	BIT1813 Controlling Whether the UGW9811 Identifies 5G Subscribers Through Automatic Learning	1	Planned on the UGW9811	This software parameter is configured using the following command: softpara bit 1813 1
RAT of subscribers	pcrf	nr	Negotiated with the peer device	These parameters are specified using the nr-rat-value command. NOTE Before the configuration, confirm the following items with the peer end: Whether the peer end can properly process the RAT value NR. Whether there is special processing for the RAT value NR. The impact brought by the addition of the RAT type must be confirmed.
	ocs	nr	Negotiated with the peer device	
	cg	nr	Negotiated with the peer device	
	aaaacct	nr	Negotiated with the peer device	
Configuring Whether to Ignore the DCNR Flag for the Specified APN	APN_BIT35	1	Planned based on the entire network	apn-softpara-bit 35

Table63

Table64 Procedure

Step 1 Enable the UGW9811 to identify 5G NSA subscribers through automatic learning.

```
softpara bit 296 1
softpara bit 327 1
softpara bit 1813 1
```

Step 2 Set the RAT values that the UGW9811 uses in messages to be sent to the CG, AAA accounting

server,PCRF, and OCS server for 5G NSA subscribers.

```
nr-rat-value pcrf xx ocs xx cg xx aaaacct xx
```

Step 3 (Optional) If differentiated charging is not required for VoLTE services, run the following command to ignore the DCNR flag for the corresponding APN (APN **ims** is used as an example):

```
apn ims
apn-softpara-bit 35 1
```

Table65 ----**End**

Table66 **Verification**

Table67 For details about how to verify the configurations, see 2.1Commissioning the 5G NSA Subscriber Information Report Feature.

Table68 **Example**

Table69 **Task Description**

Table70 Activating 4G/5G Traffic Differentiation on the UGW9811. Configure the RAT type to be reported to the OCS server for 5G NSA subscribers to **nr**.

Table71

Table72 **NOTE**

Before the configuration, confirm the following items with the peer end:

Table73 Whether the peer end can properly process the RAT value NR.

Table74 Whether there is special processing for the RAT value NR. The impact brought by the addition of the RAT type must be confirmed.

Table75 **Scripts**

1. Enable the UGW9811 to identify 5G NSA subscribers through automatic learning.

```
<UGW9811>system-view
[UGW9811]softpara bit 296 1
[UGW9811]softpara bit 327 1
[UGW9811]softpara bit 1813 1
```

2. Set the RAT value that the UGW9811 uses in messages to be sent to the OCS server for 5G NSA subscribers to **nr**.

```
<UGW9811>system-view
[UGW9811]charge-view
[UGW9811-charge] nr-rat-value ocs nr
```

Table76

3. (Optional) If differentiated charging is not required for VoLTE services, ignore the DCNR flag for the corresponding APN (APN **ims** is used as an example).

```
<UGW9811>system-view
[UGW9811]apn ims
[UGW9811-apn-test]apn-softpara-bit 35 1
```

Table77

Table78 **Example**

Table79 **Task Description**

Table80 Activating 4G/5G Traffic Differentiation on the UGW9811. Configure the RAT type to be reported to the CG, AAA accounting sever,PCRF, and OCS server for 5G NSA subscribers to **nr**.

Table81 **NOTE**

Before the configuration, confirm the following items with the peer end:

Table82 Whether the peer end can properly process the RAT value NR.

Table83 Whether there is special processing for the RAT value NR. The impact brought by the addition of the RAT type must be confirmed.

Table84 **Scripts**

1. Enable the UGW9811 to identify 5G NSA subscribers through automatic learning.

```
<UGW9811>system-view
[UGW9811]softpara bit 296 1
[UGW9811]softpara bit 327 1
[UGW9811]softpara bit 1813 1
```

2. Set the RAT value that the UGW9811 uses in messages to be sent to the CG, AAA accounting sever,PCRF, and OCS server for 5G NSA subscribers to **nr**.

```
<UGW9811>system-view
[UGW9811]charge-view
[UGW9811-charge] nr-rat-value pcrf nr ocs nr cg nr aaaacct nr
```

3. (Optional) If differentiated charging is not required for VoLTE services, ignore the DCNR flag for the corresponding APN (APN **ims** is used as an example).

```
<UGW9811>system-view
[UGW9811]apn ims
[UGW9811-apn-test]apn-softpara-bit 35 1
```

2.1 Commissioning the 5G NSA Subscriber Information Report Feature

Table85 **Scenarios**

Table86 After the 5G NSA Subscriber Information Report feature is deployed, commission this feature to ensure that it functions properly.

Table87 **NOTE**

Table88 This feature is used with the S-GW and P-GW.

Table89 **Prerequisites**

Table90 **Conditions**

Table91 Operations in 1.1 Activating the 5G NSA Subscriber Information Report Feature have been completed.

Table92 **Data**

Table93 Cat egory	Table94 Par ameter	Table95 Exa mple Value	Table96 Ho w to Obtai n	Table97 Description
Subscriber information	IMSI	460000123456789	Carried by the test MS or UE	-
APN used by the test MS or UE	APN (APN)	test	Obtained from the configured data	It is the APN configured in "Configuring a Common APN."

Table98

Table99 **Tools**

- Test MS/UE
- LMT Trace Viewer

Table100 **Procedure**

Step 1 Create a GGSN+P-GW+S-GW user trace task on the LMT, and select the default message types.

Step 2 Use the test MS/UE to request access to the network through APN **test**.

- If the test MS/UE successfully accesses the network, go to Step 3.
- If the test MS/UE fails to access the network, commission the access function of the UGW9811.

Step 3 Use the test MS/UE to browse web pages.

Step 4 View the traced messages, and check whether there are proprietary extension messages reported, as shown in Figure 1-1.

Figure 1-1 Traced messages

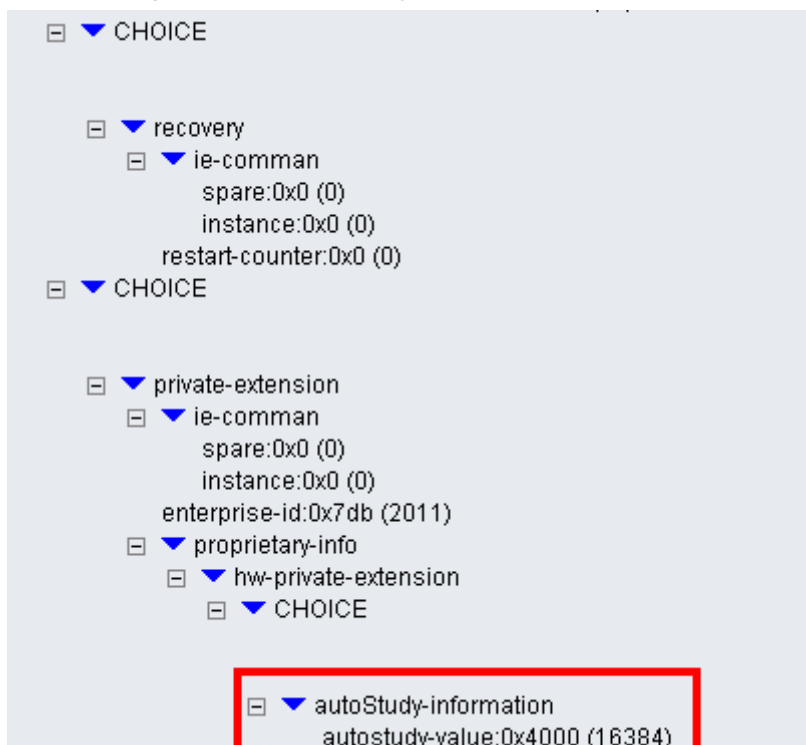


Table101

Table102

- If there are, the commissioning is complete, and no further action is required.
- If there are not, go to Step 5.

Step 5 Collect required information, and contact Huawei technical support engineers.

1. Create a user trace task on the LMT, perform Step 2, and save the traced messages.
2. Collect the above mentioned information, and contact Huawei technical support engineers.

Table103 ----**End**

2.2 Deactivating the 5G NSA Subscriber Information Report Feature

Table104 **Scenarios**

Table105 Deactivate the 5G NSA Subscriber Information Report feature on the UGW9811 if carriers no longer charge 5G NSA subscribers separately.

Table106 **NOTE**

Table107 This feature is used with the S-GW and P-GW.

Table108 **Impact on the System**

Table109 None

Table110 Prerequisites

Table111 Conditions

- Operations in 1.1 Activating the 5G NSA Subscriber Information Report Feature have been completed.
- The configuration file has been backed up locally.

Table112 Data

Table113 Category	Table114 Parameter	Table115 Example Value	Table116 How to Obtain	Table117 Description
APN	APN (APN)	test	Obtained from the configured data	It is the APN configured in "Configuring a Common APN."
5G NSA subscriber identification	BIT296 Controlling Whether the UGW9811 Updates the DCNR Flag During a Handover	1	Planned on the UGW9811	This software parameter is configured using the following command: softpara bit 296 0
	BIT327 Controlling Whether the UGW9811 Includes or Processes a DCNR or NR Flag in a Modify Bearer Request Message	1	Planned on the UGW9811	This software parameter is configured using the following command: softpara bit 327 0
	BIT1813 Controlling Whether the UGW9811 Identifies 5G Subscriber s Through	1	Planned on the UGW9811	This software parameter is configured using the following command: softpara bit 1813 0

Table113 Category	Table114 Parameter	Table115 Example Value	Table116 How to Obtain	Table117 Description
	Automatic Learning			
RAT of subscribers	pcrf	eutran	Planned on the UGW9811	These parameters are specified using the nr-rat-value command.
	ocs	eutran	Planned on the UGW9811	
	aaaacct	nr	Planned on the UGW9811	
	cg	eutran	Planned on the UGW9811	

Table118

Table119 Procedure

Step 6 Disable the UGW9811 from identifying 5G NSA subscribers through automatic learning.

```
softpara bit 296 0
softpara bit 327 0
softpara bit 1813 0
```

Step 7 Set the RAT values that the UGW9811 uses in messages to be sent to the CG, AAA accounting, PCRF, and OCS server for 5G NSA subscribers to **eutran**.

```
nr-rat-value pcrf eutran ocs eutran cg eutran aaaacct eutran
```

Table120 ----**End**

Table121 Verification

1. Run the **display softpara** command on the UGW9811 to check whether the UGW9811 identifies 5G NSA subscribers based on the DCNR flag in activation request messages.

```
[UGW9811]display softpara bit 296
Query Soft Parameter Table
-----
TYPE          ID          VALUE
BIT           296         0
(Number of results = 1)

-----
2021-06-03 14:09:08+09:00 DST

[UGW9811]display softpara bit 327
Query Soft Parameter Table
-----
```

```

TYPE          ID          VALUE
BIT           327         0
(Number of results = 1)

-----

2021-06-03 14:09:13+09:00 DST

[UGW9811]display softpara bit 1813
Query Soft Parameter Table
-----
TYPE          ID          VALUE
BIT           1813         0
(Number of results = 1)

-----

2021-06-03 14:09:19+09:00 DST
---      END

```

2. Run the **display nr-rat-value** command on the UGW9811 to check whether the RAT value in messages sent by the UGW9811 to the peripheral NEs for 5G NSA subscribers is **eutran**.

```

[UGW9811]display nr-rat-value
nr-rat-value
-----

          RAT value for PCRF = eutran
          RAT value for OCS = eutran
          RAT value for CG = eutran
RAT value for AAA Authentication = eutran
          RAT value for AAA Accounting = eutran

-----

2021-06-03 14:12:08+09:00 DST
---      END

```

Table122 Example

Table123 Task Description

Table124 Scripts

1. Disable the UGW9811 from identifying 5G NSA subscribers through automatic learning.

```

<UGW9811>system-view
[UGW9811]softpara bit 296 0
[UGW9811]softpara bit 327 0
[UGW9811]softpara bit 1813 0

```

2. Set the RAT value that the UGW9811 uses in messages to be sent to the CG, AAA accounting, PCRF, and OCS server for 5G NSA subscribers to **eutran**.

```

<UGW9811>system-view
[UGW9811]charge-view
[UGW9811-charge] nr-rat-value pcrf eutran ocs eutran cg eutran aaaacct eutran

```